AS · Unit 2 · WCH12 Energetics, Group Chemistry, Halogenoalkanes and Alcohols

Redox chemistry and groups 1, 2 and 7

Oxidation numbers turn electron transfer into simple bookkeeping, and half-equations let you build any redox equation. With those tools you can explain the chemistry of the s-block metals and the halogens: why group 2 metals get more reactive down the group while the halogens get less reactive, why carbonates become harder to decompose, and how a few test-tube reactions identify an unknown ion. The topic finishes with acid–base titrations, measurement uncertainty and the reactions of the halogens.

SPEC 8.1 Oxidation numbers and the rules

WHAT THIS SPEC POINT ASKS

Know what an oxidation number is and the rules for assigning one.

What an oxidation number is

An **oxidation number** is the charge an atom **would have if the compound were made entirely of ions**. In a covalent bond, imagine the shared electrons all go to the more electronegative atom. It is a book-keeping tool: it lets you track electrons in any reaction, ionic or covalent.

Write the sign **before** the number: +2, −1. (A charge on an ion is written the other way round: Mg^{2+}.)

The rules, in the order you apply them

Rule	Example
An **uncombined element** is **0**	Na, Fe, O_2, Cl_2, S_8 are all 0
A **simple ion**: oxidation number = its charge	Na^+ +1, Mg^{2+} +2, Cl^- −1, O^{2-} −2
The numbers in a **neutral compound add up to 0**	H_2O: 2(+1) + (−2) = 0
The numbers in an **ion add up to the ion's charge**	SO_4^{2-}: (+6) + 4(−2) = −2
Group 1 metals always **+1**; **Group 2** always **+2**	Na in NaCl, Ca in $CaCO_3$
Fluorine is always **−1**	F in HF, CaF_2, F_2O
Hydrogen is **+1**, **except in metal hydrides (−1)**	H in H_2O is +1; in NaH, CaH_2 it is −1

Rule	Example
Oxygen is **−2**, **except in peroxides (−1)** and with fluorine (+2 in F_2O)	O in H_2O_2 and Na_2O_2 is −1
Chlorine is **−1** in chlorides, but positive when bonded to O or F	Cl in ClO_3^- is +5
In a covalent bond, the **more electronegative** atom takes the **negative** value	In NH_3, N is −3 and H is +1

EXAM TIP

Fill in the "fixed" elements first (Group 1 and 2 metals, F, then H, then O), then the unknown is whatever makes the total equal the overall charge. Finally check the sum.

MARK-SCHEME WORDING

- The meaning of an oxidation number: "the **charge** the atom **would have if the species were fully ionic**" (WCH12 Jun 2024).
- ALLOW "1" or "1+" for +1 (WCH12 Oct 2020), but writing the sign first is safest.

Often asked as: give oxidation numbers in a named reaction; explain what the Roman numeral in a name means.

COMMON MISTAKE

Multiplying instead of sharing out: in H_2O each H is **+1**, not +2 (WCH12 Oct 2021 examiner report).

Giving the total for an ion (e.g. "NO_3^- changes from 5 to 4") without naming the **element** that changes (WCH12 Oct 2020 examiner report).

SPEC 8.2 Calculating oxidation numbers, including peroxides and hydrides

WHAT THIS SPEC POINT ASKS

Calculate the oxidation number of an element in compounds and ions, including peroxides and metal hydrides.

WORKED EXAMPLE

Worked examples: compounds and ions

SO_4^{2-}: S + 4(−2) = −2 → **S = +6**

$Cr_2O_7^{2-}$: 2Cr + 7(−2) = −2 → 2Cr = +12 → **Cr = +6**

MnO_4^-: Mn + 4(−2) = −1 → **Mn = +7**

NH_4^+: N + 4(+1) = +1 → **N = −3**

ClO_3^-: Cl + 3(−2) = −1 → **Cl = +5**

$KClO_4$: (+1) + Cl + 4(−2) = 0 → **Cl = +7**

$AgNO_3$: Ag^+ is +1 (it's the ion Ag^+ and NO_3^-); N + 3(−2) = −1 → **N = +5**

IO_3^-: I + 3(−2) = −1 → **I = +5**

Working out oxidation numbers

1. **Iron in $FeCl_3$.** Each chlorine is −1, so three chlorines give −3. The compound is neutral, so iron must be **+3**.

   The oxidation number is for one atom of the element. Chlorine in $CaCl_2$ is −1, not −2.

2. **Sulfur in SO_4^{2-}.** Let sulfur be x. Four oxygens at −2 each, and the total must equal the charge on the ion:

$$x + 4(-2) = -2 \quad \Rightarrow \quad x = +6$$

3. **Chromium in $K_2Cr_2O_7$.** Potassium is +1 and there are two chromium atoms:

$$2(+1) + 2x + 7(-2) = 0 \quad \Rightarrow \quad 2x = +12 \quad \Rightarrow \quad x = +6$$

4. **Nitrogen in NH_4^+.** $x + 4(+1) = +1$, so nitrogen is **−3**.

WORKED EXAMPLE

Worked examples: the exceptions

H_2O_2 (hydrogen peroxide): 2(+1) + 2O = 0 → **O = −1**

Na_2O_2 (sodium peroxide): 2(+1) + 2O = 0 → **O = −1**

BaO_2 (barium peroxide): (+2) + 2O = 0 → **O = −1**

NaH (sodium hydride): Na is +1, so **H = −1**

CaH_2 (calcium hydride): (+2) + 2H = 0 → **H = −1**

$LiAlH_4$: Li +1, Al +3, so 4H = −4 → **H = −1**

How to spot them: a peroxide has an O–O bond (the formula has "too many" oxygens for O to be −2); a metal hydride is a metal joined to H only.

Fractional and average values

- In **$S_4O_6^{2-}$** (tetrathionate), 4S + 6(−2) = −2 → 4S = +10 → S = **+2.5**. That is an **average**: the sulfur atoms are actually in different environments.
- In **Fe_3O_4**, Fe averages **+⅔** (it contains both Fe^{2+} and Fe^{3+}).

MARK-SCHEME WORDING

- Iodine species: "IO_3^- = +5, I^- = −1, I_2 = 0". DO NOT AWARD I^- = −5 (WCH12 Jun 2022).
- Manganese and carbon in MnO_4^- and $C_2O_4^{2-}$: Mn +7, C +3 (WCH12 Jan 2025).

Often asked as: oxidation number of an element in an unfamiliar ion; changes during a reaction.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Sulfur: **−2** in H_2S, **0** in S, **+4** in SO_2, **+6** in H_2SO_4 and SO_4^{2-}. You need these for the halide + concentrated sulfuric acid reactions (WCH12 Jun 2022).

Cobalt goes from +4 in CoO_2 to +3 in $LiCoO_2$ in a lithium-ion cell (WCH12 Oct 2024).

COMMON MISTAKE

Silver as +9 in $AgNO_3$: the +1 belongs to Ag, and the rest of the charge belongs to the nitrate ion (WCH12 Oct 2020 examiner report).

Chlorine as −2 in $CaCl_2$ (it's −1: two chlorines share the −2 total) (WCH12 Jan 2023 examiner report).

Forgetting that H is −1 in a metal hydride and O is −1 in a peroxide.

SPEC 8.3 Oxidation numbers as Roman numerals

WHAT THIS SPEC POINT ASKS

Show the oxidation number of an element in a compound or ion with a Roman numeral.

When and how

- Use a Roman numeral when an element can have **more than one** oxidation number: transition metals, and non-metals in oxyanions.
- The numeral goes in brackets **straight after the element it describes**, with no space: iron(III) chloride.
- For a negative ion ending in **-ate**, the numeral describes the central element: sulfate(VI), chlorate(V).
- Group 1 and 2 metals never need one: it's sodium chloride, not sodium(I) chloride.

Name	Formula	Element and its oxidation number
iron(II) sulfate	$FeSO_4$	Fe +2
iron(III) chloride	$FeCl_3$	Fe +3
copper(I) oxide	Cu_2O	Cu +1
manganese(IV) oxide	MnO_2	Mn +4
sodium chlorate(I)	$NaClO$	Cl +1 (the active ingredient in bleach)
sodium chlorate(V)	$NaClO_3$	Cl +5
potassium chlorate(VII)	$KClO_4$	Cl +7
potassium manganate(VII)	$KMnO_4$	Mn +7
potassium dichromate(VI)	$K_2Cr_2O_7$	Cr +6
sulfate(VI) / sulfate(IV)	SO_4^{2-} / SO_3^{2-}	S +6 / +4
nitrate(V) / nitrate(III)	NO_3^- / NO_2^-	N +5 / +3

Everyday names drop the numeral when there's no doubt: "sulfate" means sulfate(VI), "nitrate" means nitrate(V), "nitrite" is nitrate(III).

MARK-SCHEME WORDING

- "(V) is the **oxidation number of the chlorine** (in the ion)" or "the charge chlorine would have if the ion were fully ionic". DO NOT AWARD "the charge on chlorate(V) is +5" (WCH12 Jun 2024).

- "Potassium chlorate(VII)" is $KClO_4$; "potassium chlorate" alone is not enough (WCH12 Jun 2021).

Often asked as: name a compound with its numeral; explain what the numeral means.

COMMON MISTAKE

Saying the numeral is the charge on the **ion**. Chlorate(V), ClO_3^-, has a charge of −1; the V refers to the chlorine.

SPEC 8.4 Writing formulae from oxidation numbers

WHAT THIS SPEC POINT ASKS

Write the formula of a compound or ion when given its name with oxidation numbers.

Method

1. Write the oxidation number of each element (from the numeral, or the fixed rules).
2. For an oxyanion, work out how many O atoms give the right charge; for a compound, balance the total to zero.
3. Use brackets round a polyatomic ion if you need more than one.

WORKED EXAMPLE

Worked examples

Tin(IV) oxide: Sn +4, O −2 → need 2 O → **SnO_2**

Iron(III) sulfate: Fe^{3+} and SO_4^{2-} → lowest common multiple 6 → **$Fe_2(SO_4)_3$**

Lead(II) nitrate: Pb^{2+} and NO_3^- → **$Pb(NO_3)_2$**

Sodium chlorate(V): Cl +5 with O −2. For an ion of charge −1: +5 + 3(−2) = −1 → ClO_3^- → **$NaClO_3$**

Potassium chromate(VI): Cr +6; +6 + 4(−2) = −2 → CrO_4^{2-} → **K_2CrO_4**

Sodium bromate(I): Br +1; +1 + (−2) = −1 → BrO^- → **NaBrO**

EXAM TIP

With an oxyanion, the usual charges are −1 (ClO_3^-, NO_3^-, MnO_4^-) or −2 (SO_4^{2-}, CO_3^{2-}, CrO_4^{2-}). Choose the number of oxygens that gives one of these, then check.

COMMON MISTAKE

Writing ClO_5 for chlorate(V). The V is the oxidation number, not the number of oxygens.

SPEC 8.5 Oxidation and reduction in s- and p-block reactions

WHAT THIS SPEC POINT ASKS

Understand oxidation and reduction as electron transfer and as changes in oxidation number, and apply this to reactions of s-block and p-block elements.

Two ways to spot redox

	Oxidation	Reduction
Electrons	**loss** of electrons	**gain** of electrons
Oxidation number	**increases**	**decreases**

Memory aid: **OIL RIG** (Oxidation Is Loss, Reduction Is Gain). A redox reaction always has both: electrons lost by one species are gained by another.

s-block and p-block examples

Reaction	Oxidised	Reduced
$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$	Na: 0 → +1	Cl: 0 → −1
$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$	Mg: 0 → +2	O: 0 → −2
$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$	Mg: 0 → +2	H: +1 → 0
$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$	K: 0 → +1	H: +1 → 0
$Cl_2(aq) + 2KBr(aq) \rightarrow 2KCl(aq) + Br_2(aq)$	Br: −1 → 0	Cl: 0 → −1
$Ag(s) + 2HNO_3(aq) \rightarrow AgNO_3(aq) + NO_2(g) + H_2O(l)$	Ag: 0 → +1	N: +5 → +4

EXAM TIP

Write the oxidation number above **every** element first. Usually only two elements change. Name the **element**, give both numbers ("from 0 to +1") and say which is oxidised and which reduced.

MARK-SCHEME WORDING

- "Silver is **oxidised from 0 to +1**; nitrogen is **reduced from +5 to +4**" (WCH12 Oct 2020).
- For K + water: "potassium oxidised 0 → +1; **hydrogen** reduced +1 → 0" (WCH12 Oct 2021).
- When the question says "in terms of oxidation numbers", answer with numbers, not electrons.

Often asked as: identify what is oxidised and reduced in a given equation, with oxidation numbers.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Redox in industrial and environmental contexts: in selective catalytic reduction, NH_3 is oxidised (N −3 → 0) and NO reduced (N +2 → 0) (WCH12 Jan 2020).

COMMON MISTAKE

Hydrogen "starting at +2" in water because there are two of them; naming water or oxygen as the species reduced in K + H_2O (WCH12 Oct 2021 examiner report).

Showing a change for a spectator such as calcium (WCH12 Jan 2023 examiner report).

Swapping oxidation and reduction.

SPEC 8.6 Oxidising agents and reducing agents

WHAT THIS SPEC POINT ASKS

Know that oxidising agents gain electrons and reducing agents lose electrons.

The definitions

- An **oxidising agent accepts (gains) electrons** from another species, so it oxidises that species. The oxidising agent is itself **reduced** (its oxidation number falls).
- A **reducing agent donates (loses) electrons**, so it reduces another species. The reducing agent is itself **oxidised** (its oxidation number rises).

In $2Na + Cl_2 \rightarrow 2NaCl$: Cl_2 is the oxidising agent (gains electrons); Na is the reducing agent (loses electrons).

Common **oxidising agents**	Common **reducing agents**
halogens (F_2 > Cl_2 > Br_2 > I_2)	s-block metals (Group 1, Group 2) and other reactive metals
O_2	H_2, C, CO
acidified MnO_4^- (manganate(VII))	halide ions (I^- > Br^- > Cl^-)
acidified $Cr_2O_7^{2-}$ (dichromate(VI))	Fe^{2+}, SO_2
concentrated H_2SO_4, HNO_3	
H_2O_2	H_2O_2 (with a stronger oxidising agent)

- **Hydrogen peroxide** can be both. As an oxidant, O goes −1 → −2 (e.g. with I^- or Fe^{2+} in acid). As a reductant, O goes −1 → 0 and O_2 is given off (e.g. with acidified MnO_4^-).

MARK-SCHEME WORDING

- "Iodide ions are **better reducing agents** than chloride ions" because "**electrons are more easily lost** from iodide ions" (the outer electrons are further from the nucleus and more shielded) (WCH12 Jun 2025).
- DO NOT AWARD "iodine" or "chlorine" for the ions, or anything linking it to a better **oxidising** agent.

Often asked as: name the oxidising or reducing agent in an equation; compare the reducing power of two halide ions.

COMMON MISTAKE

Saying the oxidising agent is oxidised. It is the opposite: it **causes** oxidation and is itself reduced.

Writing "iodine is a stronger reducing agent" when you mean **iodide** ions.

SPEC 8.7 Disproportionation

WHAT THIS SPEC POINT ASKS

Understand that in disproportionation an element in a single species is oxidised and reduced at the same time.

The definition

Disproportionation: a reaction in which **the same element in the same species** is **simultaneously oxidised and reduced**. One species goes in; the element comes out in two different oxidation states, one higher and one lower.

Reaction	Oxidation numbers of the element
$Cl_2 + H_2O \rightleftharpoons HCl + HClO$	Cl: 0 → −1 (reduced) and 0 → +1 (oxidised)
$Cl_2 + 2NaOH \rightarrow NaCl + NaClO + H_2O$ (cold, dilute)	Cl: 0 → −1 and 0 → +1
$3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$ (hot)	Cl: 0 → −1 and 0 → +5
$2H_2O_2 \rightarrow 2H_2O + O_2$	O: −1 → −2 and −1 → 0
$2Cu^+(aq) \rightarrow Cu(s) + Cu^{2+}(aq)$	Cu: +1 → 0 and +1 → +2
$3ClO^- \rightarrow 2Cl^- + ClO_3^-$ (chlorate(I) on heating)	Cl: +1 → −1 and +1 → +5

WORKED EXAMPLE

Worked example: answering a disproportionation question

"Explain why $Cl_2 + 2OH^- \rightarrow Cl^- + ClO^- + H_2O$ is a disproportionation."

- It is a **disproportionation** reaction.
- Chlorine is **oxidised** from **0** in Cl_2 to **+1** in ClO^-.
- Chlorine is **reduced** from **0** in Cl_2 to **−1** in Cl^-.
- The same element in one species (Cl_2) is oxidised and reduced at the same time.

MARK-SCHEME WORDING

- "disproportionation" (a stand-alone mark), then "chlorine **oxidised** from 0 to +1 (or +5)" and "**reduced** from 0 to −1". "Reference to 0 for Cl_2 required once only" (WCH12 Jun 2021, Oct 2025).
- Oxidation numbers may be written on the equation; three correct numbers without the oxidised/reduced links score one mark.
- Do **not** give changes for H or O: "Do not award … if changes of oxidation numbers of H and O given" (WCH12 Jun 2023).

Often asked as: Cl_2 with water, cold NaOH, hot NaOH or $Ca(OH)_2$; a 6-mark question on all three (WCH12 Jan 2022).

COMMON MISTAKE

Not saying it is **simultaneous** oxidation and reduction, or not linking "oxidised" to an increase and "reduced" to a decrease (WCH12 Jan 2023 examiner report).

Adding oxidation-number changes for H, O, Na or Ca, which don't change.

SPEC 8.8 Using oxidation numbers to classify reactions

WHAT THIS SPEC POINT ASKS

Know that oxidation numbers let you classify a reaction as redox and as disproportionation.

The decision tree

1. Write oxidation numbers for every element on both sides.
2. **No element changes** → **not redox** (e.g. acid–base, precipitation, many thermal decompositions).
3. One element goes up and another goes down → **redox**.
4. The **same element in one species** goes both up and down → **disproportionation** (a special kind of redox).

Reaction	Changes?	Type
$NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$	none	acid–base, **not redox**
$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$	none	precipitation, **not redox**
$CaCO_3 \rightarrow CaO + CO_2$	none	thermal decomposition, **not redox**
$2NaBr + 2H_2SO_4 \rightarrow Br_2 + SO_2 + Na_2SO_4 + 2H_2O$	Br −1 → 0; S +6 → +4	**redox**
$2Mg(NO_3)_2 \rightarrow 2MgO + 4NO_2 + O_2$	N +5 → +4; O −2 → 0	**redox** (but not disproportionation: two different elements)
$IO_3^- + 5I^- + 6H^+ \rightarrow 3I_2 + 3H_2O$	I +5 → 0 and I −1 → 0	**redox, not disproportionation**: two different species of iodine end up in the same state

MARK-SCHEME WORDING

- Not redox: "**no oxidation number has changed**" (ALLOW "no electrons transferred") (WCH12 Jun 2025).
- IO_3^- + I^-: not disproportionation because "**two different species** (of iodine) are oxidised and reduced (to form the same species)" / "there is not one species being oxidised and reduced" (WCH12 Jun 2022).

Often asked as: show that a reaction is (or is not) redox or disproportionation.

COMMON MISTAKE

Writing "compounds" when you mean species or ions (WCH12 Jun 2022 examiner report).

Saying both species are reduced in the iodate–iodide reaction.

SPEC 8.9 Metals form positive ions; non-metals form negative ions

WHAT THIS SPEC POINT ASKS

Understand that metals generally lose electrons to form positive ions (oxidation number rises), and non-metals generally gain electrons to form negative ions (oxidation number falls).

The pattern

- **Metals** have few outer electrons and relatively low ionisation energies. They **lose** electrons to form **positive ions**: $Na \rightarrow Na^+ + e^-$, $Ca \rightarrow Ca^{2+} + 2e^-$. Their oxidation number **increases**, so metals are **oxidised** and act as **reducing agents**.
- **Non-metals** have nearly full outer shells and attract electrons strongly. They **gain** electrons to form **negative ions**: $Cl_2 + 2e^- \rightarrow 2Cl^-$, $O_2 + 4e^- \rightarrow 2O^{2-}$. Their oxidation number **decreases**, so non-metals are **reduced** and act as **oxidising agents**.
- This is why Group 1 and 2 metals are reducing agents (8.12) and the halogens are oxidising agents (8.24).

EXAM TIP

A metal is never found with a negative oxidation number in a simple compound. If you get one, recheck.

COMMON MISTAKE

Saying "non-metals lose electrons" by analogy with metals. Non-metals (other than H when it bonds to a more electronegative atom) gain electrons when they react with metals.

SPEC 8.10 Ionic half-equations and full ionic equations

WHAT THIS SPEC POINT ASKS

Write ionic half-equations and combine them into a full ionic equation.

Writing a half-equation (acidic solution)

1. Write the species that changes and what it becomes. **Balance that element.**
2. Balance **O** by adding **H_2O**.
3. Balance **H** by adding **H^+**.
4. Balance **charge** by adding **electrons** (e^-) to the more positive side.
5. Check: atoms and total charge both balance.

Worked example: manganate(VII) to manganese(II), which gives **$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$** (reduction: electrons on the left).

Half-equation for $MnO_4^- \rightarrow Mn^{2+}$

1 Balance the atoms of the element that changes (here one Mn on each side is already balanced), and find its change in oxidation number: Mn goes from +7 to +2, a gain of 5 electrons.

$$MnO_4^- + 5e^- \rightarrow Mn^{2+}$$

2 Balance oxygen by adding water to the side that needs it. Four O atoms on the left need four H_2O on the right:

$$MnO_4^- + 5e^- \rightarrow Mn^{2+} + 4H_2O$$

3 Balance hydrogen by adding H^+ ions. Eight H atoms in the water need $8H^+$ on the left:

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$$

4 Check that the charges balance. Left: (−1) + (+8) + (−5) = +2. Right: +2. Balanced.

If you only balance the atoms and not the charges, the number of electrons is probably wrong.

Half-equation for $SO_4^{2-} \rightarrow H_2S$

1. Sulfur goes from +6 to −2, a gain of 8 electrons:

$$\mathrm{SO_4^{2-} + 8e^- \rightarrow H_2S}$$

2. Four O atoms on the left, so add $4H_2O$ on the right:

$$\mathrm{SO_4^{2-} + 8e^- \rightarrow H_2S + 4H_2O}$$

3. There are now 2 + 8 = 10 H atoms on the right, so add $10H^+$ on the left:

$$\mathrm{SO_4^{2-} + 10H^+ + 8e^- \rightarrow H_2S + 4H_2O}$$

4. Check the charges. Left: (−2) + (+10) + (−8) = 0. Right: 0. Balanced.

Combining two half-equations

1. One half-equation is a reduction (electrons on the **left**); the other an oxidation (electrons on the **right**).
2. Multiply one or both so the **number of electrons is the same**.
3. Add them and **cancel** the electrons and anything else that appears on both sides (H^+, H_2O).

WORKED EXAMPLE

Worked example: hydrogen peroxide oxidises iodide

Oxidation: $2I^- \rightarrow I_2 + 2e^-$

Reduction: $H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$

Electrons already equal (2), so add:

$H_2O_2 + 2H^+ + 2I^- \rightarrow I_2 + 2H_2O$

Worked example: manganate(VII) oxidises iron(II). Oxidation: $Fe^{2+} \rightarrow Fe^{3+} + e^-$ (× 5). Reduction: $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$. Overall: **$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$**. Check charge: left −1 + 8 + 10 = +17; right +2 + 15 = +17 ✓

Manganate(VII) ions oxidising iron(II) ions

1. The two half-equations:

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$$

$$Fe^{2+} \rightarrow Fe^{3+} + e^-$$

2. The reduction takes 5 electrons and the oxidation gives 1, so multiply the iron half-equation by 5:

$$5Fe^{2+} \rightarrow 5Fe^{3+} + 5e^-$$

3. Add the two and cancel the $5e^-$:

$$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$$

Check: charge on the left = −1 + 8 + 10 = +17; on the right = +2 + 15 = +17.

WORKED EXAMPLE

Worked example: chlorine in cold alkali

In alkali, balance with OH^- and H_2O instead of H^+.

Reduction: $Cl_2 + 2e^- \rightarrow 2Cl^-$

Oxidation: $Cl_2 + 4OH^- \rightarrow 2ClO^- + 2H_2O + 2e^-$

Add and divide by 2: **$Cl_2 + 2OH^- \rightarrow Cl^- + ClO^- + H_2O$**

Full ionic equations without half-equations

Write the full equation with state symbols, split soluble ionic compounds into ions, then **cancel the spectator ions** (ions unchanged on both sides).

- Precipitation: $BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$ becomes **$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$**
- Acid + carbonate: **$CO_3^{2-} + 2H^+ \rightarrow CO_2 + H_2O$**
- Displacement: $Cl_2 + 2KBr \rightarrow 2KCl + Br_2$ becomes **$Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$** ($K^+$ is a spectator)

MARK-SCHEME WORDING

- $H_2O_2 + I^-$: "$2I^- \rightarrow I_2 + 2e^-$" and "$H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$"; overall "$2I^- + 2H^+ + H_2O_2 \rightarrow I_2 + 2H_2O$". "**Do not award uncancelled electrons**" (WCH12 Jun 2019).
- "$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$"; "$C_2H_2O_4 \rightarrow 2CO_2 + 2H^+ + 2e^-$" (WCH12 Jan 2025).

- Barium sulfate: "$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$", **state symbols needed**; DO NOT AWARD with spectator ions (WCH12 Oct 2024).
- Carbonate + acid: "$CO_3^{2-} + 2H^+ \rightarrow CO_2 + H_2O$"; DO NOT AWARD H_2CO_3 as the product (WCH12 Oct 2019).
- Multiples of a correct equation are allowed.

Often asked as: combine two given half-equations; write the half-equation for an unfamiliar species; ionic equations for precipitation, neutralisation and displacement.

COMMON MISTAKE

Putting O^{2-} ions in an acidic half-equation: use H_2O and H^+ (WCH12 Jun 2019 examiner report).

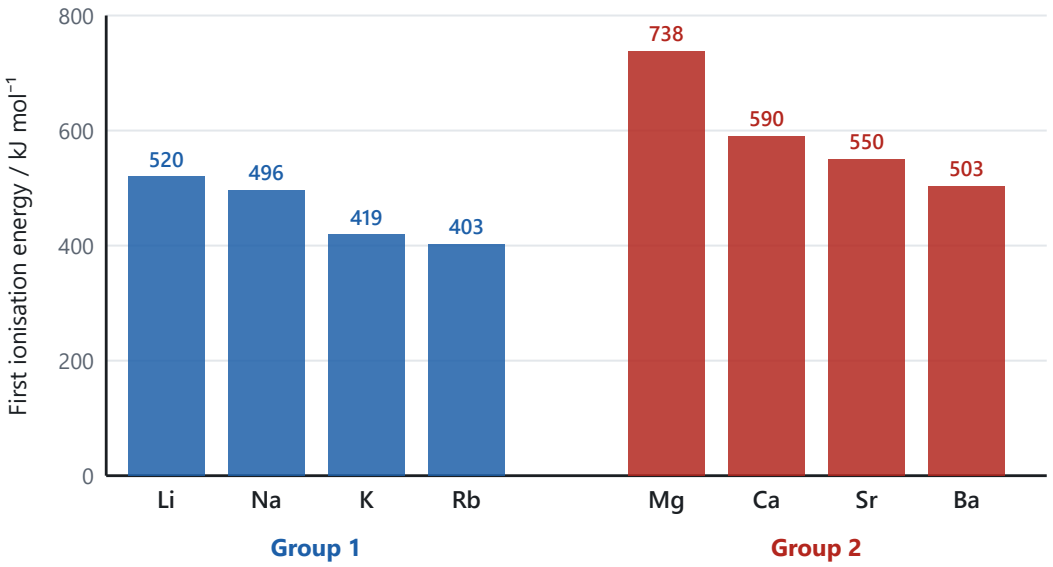
Both half-equations with electrons on the same side. One must be a reduction and one an oxidation.

Leaving electrons or spectator ions in the final equation; wrong charges (CO_3^- instead of CO_3^{2-}).

SPEC 8.11 Ionisation energy down Groups 1 and 2

WHAT THIS SPEC POINT ASKS

Understand the reasons for the trend in ionisation energy down Groups 1 and 2.



First ionisation energy falls down both groups. Each Group 2 element has a higher value than the Group 1 element in the same period.

The trend and the reason

First ionisation energy decreases down both groups. Going down:

1. Each element has **one more occupied shell**, so the outer electron is **further from the nucleus** (larger atomic radius).
2. There are **more inner shells shielding** the outer electron from the nucleus.
3. The nuclear charge does increase, but the extra distance and shielding outweigh it. The **attraction between the nucleus and the outer electron gets weaker**, so **less energy is needed to remove it**.

Group 2 needs two electrons removed

- Group 2 metals form M^{2+}, so what matters for reactivity is the **sum of the first and second ionisation energies**. That sum also falls down the group: about 2190 kJ mol^{-1} for Mg down to about 1470 for Ba.
- The **second** ionisation energy is always bigger than the first: the electron is removed from a **positive ion**, so it is held more strongly (same shell, but the ion is smaller and the remaining electrons are pulled closer).
- Group 2 values are higher than Group 1 in the same period because the Group 2 atom has **one more proton** in the same shell, so a stronger attraction.

MARK-SCHEME WORDING

- "Atomic radius increases / **more shells** / **more shielding**, so the outer electron is **less strongly attracted** (to the nucleus) and **easier to remove**".
- DO NOT AWARD "nuclear charge decreases". ALLOW "effective nuclear charge decreases" (WCH12 Jun 2022, by analogy with electronegativity).

Often asked as: rarely on its own; it is the first step of a reactivity explanation (8.12).

COMMON MISTAKE

Saying the nuclear charge falls down a group. It rises; shielding and distance win.

Using ionisation energy to explain thermal stability of carbonates and nitrates. That needs polarisation (8.16) (WCH12 Jan 2023 examiner report).

SPEC 8.12 Reactivity trends in Groups 1 and 2

WHAT THIS SPEC POINT ASKS

Understand why reactivity increases down Group 1 (Li to K) and Group 2 (Mg to Ba).

The explanation

- When a Group 1 or 2 metal reacts, it **loses its outer electron(s)** to form a positive ion. It is acting as a **reducing agent**.
- Down each group, the outer electrons are **further from the nucleus** and **more shielded**, so they are **lost more easily** (lower ionisation energy, 8.11).
- So reactivity **increases down the group**: K > Na > Li, and Ba > Sr > Ca > Mg.
- Group 1 metals are **more reactive** than the Group 2 metal in the same period: they need to lose only **one** electron, and their ionisation energy is lower.

EXAM TIP

A full answer: "Down the group the atoms have more shells, so the outer electron is further from the nucleus and more shielded. The attraction is weaker, so the electron is lost more easily, and the metal reacts more vigorously."

COMMON MISTAKE

"The atoms get bigger so they are more reactive", with no link to losing electrons.

Saying the metal **gains** electrons when it reacts.

SPEC 8.13 Reactions of Group 1 and 2 elements with oxygen, chlorine and water

WHAT THIS SPEC POINT ASKS

Know the reactions of Group 1 (Li to K) and Group 2 (Mg to Ba) with oxygen, chlorine and water.

With oxygen

Metal	Equation (normal oxide)	Observations
Li	$4Li(s) + O_2(g) \rightarrow 2Li_2O(s)$	burns with a **red** flame, white solid
Na	$4Na(s) + O_2(g) \rightarrow 2Na_2O(s)$	burns with a **yellow-orange** flame
K	$4K(s) + O_2(g) \rightarrow 2K_2O(s)$	burns with a **lilac** flame
Mg	$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$	**bright white** flame, white powder
Ca	$2Ca(s) + O_2(g) \rightarrow 2CaO(s)$	**brick-red** (orange-red) flame, white solid

Metal	Equation (normal oxide)	Observations
Sr, Ba	$2M(s) + O_2(g) \rightarrow 2MO(s)$	red (Sr) and pale green (Ba) flames

- Group 1 metals **tarnish** quickly in air, faster down the group, so they are stored **under oil**.
- In excess oxygen, sodium mainly forms the **peroxide** Na_2O_2 and potassium the **superoxide** KO_2; barium can form BaO_2. The simple oxides above are what you'd normally be asked for.

With chlorine

All react on heating to give white ionic chlorides, more vigorously down each group:

- Group 1: **$2M(s) + Cl_2(g) \rightarrow 2MCl(s)$**, e.g. $2Na + Cl_2 \rightarrow 2NaCl$ (sodium burns with a yellow flame in chlorine).
- Group 2: **$M(s) + Cl_2(g) \rightarrow MCl_2(s)$**, e.g. $Mg + Cl_2 \rightarrow MgCl_2$.

With water

Group 1: $2M(s) + 2H_2O(l) \rightarrow 2MOH(aq) + H_2(g)$. The solution is strongly alkaline.

Metal	What you see
Li	floats, **fizzes** steadily, moves slowly, gradually **disappears**
Na	floats, **melts into a silvery ball** (low melting point, exothermic reaction), whizzes around, fizzes, disappears
K	as sodium but more violent; the hydrogen **ignites with a lilac flame**; may spit

Group 2: $M(s) + 2H_2O(l) \rightarrow M(OH)_2(aq \text{ or } s) + H_2(g)$

Metal	What you see
Mg	**very slow** with cold water (a few bubbles over days). With **steam** it burns with a white light: **$Mg(s) + H_2O(g) \rightarrow MgO(s) + H_2(g)$**
Ca	**fizzes steadily**, the water turns **cloudy white** ($Ca(OH)_2$ doesn't dissolve), the mixture warms
Sr, Ba	faster and more vigorous; less cloudiness down the group because the hydroxides get more soluble

MARK-SCHEME WORDING

- Ca + water: "$Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$" (ALLOW $CaO + H_2$). The H_2 is often forgotten (WCH12 Oct 2019).

- K + water: "$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$", **state symbols marked** (WCH12 Oct 2021). Li: "$Li(s) + H_2O(l) \rightarrow LiOH(aq) + \frac{1}{2}H_2(g)$" (WCH12 Oct 2024).
- Observations: "bubbles / fizzing / effervescence", "lithium disappears / gets smaller", "floats / moves on the surface". IGNORE "hydrogen produced", "exothermic"; DO NOT AWARD "white precipitate" or "smoke" for lithium (WCH12 Oct 2024).
- Ca + O_2: "$2Ca(s) + O_2(g) \rightarrow 2CaO(s)$" (WCH12 Oct 2025).

Often asked as: equation with state symbols and observations for a named metal and water; pH of the solution formed.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Lithium also reacts with **nitrogen** in air to form lithium nitride, **Li_3N** ($6Li + N_2 \rightarrow 2Li_3N$). Very few candidates could write it (WCH12 Oct 2024).

Identifying a Group 1 metal from a titration of its hydroxide: find moles of MOH, then A_r = mass ÷ moles (WCH12 Oct 2021).

COMMON MISTAKE

"Hydrogen was formed" as an observation: you see **bubbles**, you don't see hydrogen (WCH12 Oct 2024 examiner report).

$LiOH(s)$ or $H_2O(aq)$ as state symbols.

Forgetting H_2 in the Group 2 equation.

SPEC 8.14 Reactions of oxides and hydroxides with water and acid

WHAT THIS SPEC POINT ASKS

Know the reactions of (i) Group 1 and 2 oxides with water and dilute acid and (ii) Group 1 and 2 hydroxides with dilute acid.

(i) Oxides with water

The oxide ion reacts with water to make hydroxide ions: **$O^{2-} + H_2O \rightarrow 2OH^-$**. So the solutions are **alkaline**.

Oxide	Equation	pH of the result
Na_2O	$Na_2O(s) + H_2O(l) \rightarrow 2NaOH(aq)$	about 14
Li_2O, K_2O	similar: $M_2O + H_2O \rightarrow 2MOH$	about 14
MgO	$MgO(s) + H_2O(l) \rightarrow Mg(OH)_2(s)$ (only slightly)	about 9–10: $Mg(OH)_2$ is barely soluble
CaO	$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq/s)$ (very exothermic: "slaking" lime)	about 12
BaO	$BaO(s) + H_2O(l) \rightarrow Ba(OH)_2(aq)$	about 13

(i) Oxides with dilute acid

Oxides are **bases**: they neutralise acids to give a **salt and water**.

- $MgO(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2O(l)$; ionic: $MgO(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2O(l)$
- $Na_2O(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l)$
- $CaO(s) + 2HNO_3(aq) \rightarrow Ca(NO_3)_2(aq) + H_2O(l)$

(ii) Hydroxides with dilute acid

$OH^-(aq) + H^+(aq) \rightarrow H_2O(l)$ in every case.

- $NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$
- $Ca(OH)_2(aq) + 2HCl(aq) \rightarrow CaCl_2(aq) + 2H_2O(l)$
- $Mg(OH)_2(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + 2H_2O(l)$ ($Mg(OH)_2$ in **antacids** neutralises excess stomach acid)

With **sulfuric acid**, calcium, strontium and especially barium compounds form a layer of **insoluble sulfate** on the solid, which slows or stops the reaction.

WORKED EXAMPLE

Worked example: an antacid

An antacid tablet neutralises 0.0240 mol of HCl. What mass of $Mg(OH)_2$ (M_r 58.3) does it contain?

$Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$, so $n(Mg(OH)_2) = 0.0240 \div 2 = 0.0120$ mol

mass = 0.0120×58.3 = **0.700 g**

MARK-SCHEME WORDING

- Mass from moles: "0.150 ÷ 2 = 0.075 mol", "M_r 58.3", "4.37 g" (WCH12 Oct 2019).
- $Ca(OH)_2$ solubility by titration: moles of HCl in the titre, ÷ 2, scale to 1 dm^3 (× 1000 ÷ 25.0), × 74.1 → g dm^{-3} (WCH12 Oct 2020).

Often asked as: equations with acid; titration calculations for $Ca(OH)_2$ or CaO; back titration for the purity of a solid.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Back titration for the purity of CaO: add a known excess of HCl, titrate the leftover acid with NaOH, subtract to find the acid used by CaO, ÷ 2, then mass and % purity (WCH12 Oct 2025, 6 marks).

COMMON MISTAKE

Forgetting the 1 : 2 ratio for $Ca(OH)_2$ or MgO with HCl (WCH12 Oct 2020 examiner report).

Writing O^{2-} ions in an equation for a reaction in acid (use the solid oxide or H_2O).

SPEC 8.15 Solubility of Group 2 hydroxides and sulfates

WHAT THIS SPEC POINT ASKS

Know the trends in solubility of the Group 2 hydroxides and sulfates.

Opposite trends

	Mg	Ca	Sr	Ba
Hydroxide, $M(OH)_2$	very sparingly soluble	sparingly soluble	fairly soluble	**soluble**
Sulfate, MSO_4	**soluble**	sparingly soluble	very sparingly soluble	**insoluble**

- **Hydroxides become more soluble down the group.** So the solutions contain more OH^- and are **more alkaline** further down: $Mg(OH)_2$ suspension about pH 10, saturated $Ba(OH)_2$ about pH 13+.
- **Sulfates become less soluble down the group.** $BaSO_4$ is effectively insoluble.

Uses that follow from the trends

- **$Mg(OH)_2$** ("milk of magnesia"): an antacid and laxative. Low solubility means it's only mildly alkaline, so safe to swallow.
- **$Ca(OH)_2$** (slaked lime): neutralises acidic soils and lakes; its saturated solution is **limewater**.
- **$BaSO_4$**: the "barium meal" for X-rays of the gut. Barium ions are **toxic**, but $BaSO_4$ is so insoluble that almost no Ba^{2+} is absorbed.
- **$BaCl_2(aq)$**: the test for sulfate ions (8.19), because $BaSO_4$ precipitates.

MARK-SCHEME WORDING

- "Calcium hydroxide is **more soluble than** magnesium hydroxide" / "solubility of the hydroxides increases down the group", so the "**concentration of hydroxide ions is greater**" (WCH12 Oct 2019).
- With $Mg(OH)_2$ instead of $Ca(OH)_2$, "magnesium hydroxide is less soluble, so the OH^- concentration is lower and the titre would be smaller" (WCH12 Oct 2020). DO NOT AWARD "magnesium hydroxide is insoluble".
- Barium meal: "barium ions are precipitated as (insoluble) barium sulfate" (WCH12 Oct 2024).

Often asked as: compare the pH of two Group 2 hydroxide solutions; predict a titre; explain the barium meal.

COMMON MISTAKE

Mixing up the directions: it's hydroxides **up** in solubility, sulfates **down**. Memory aid: "**Ba**se gets **better** (more soluble hydroxide), **Ba**SO_4 is **bad** (insoluble)."

A vague pH statement with no link to OH^- concentration or solubility (WCH12 Oct 2019 examiner report).

SPEC 8.16 Thermal stability of nitrates and carbonates

WHAT THIS SPEC POINT ASKS

Understand the trends in thermal stability of Group 1 and 2 nitrates and carbonates in terms of the size and charge of the cations.

What happens on heating

Compound	Decomposition	Example equation
Group 2 carbonates	→ oxide + CO_2	$MgCO_3(s) \rightarrow MgO(s) + CO_2(g)$

Compound	Decomposition	Example equation
Li_2CO_3	→ oxide + CO_2 (the only Group 1 carbonate that decomposes at Bunsen temperatures)	$Li_2CO_3(s) \rightarrow Li_2O(s) + CO_2(g)$
Na_2CO_3, K_2CO_3	**no decomposition** at Bunsen temperatures	—
Group 2 nitrates	→ oxide + **NO_2** + O_2	$2Mg(NO_3)_2(s) \rightarrow 2MgO(s) + 4NO_2(g) + O_2(g)$
$LiNO_3$	→ oxide + NO_2 + O_2 (like Group 2)	$4LiNO_3(s) \rightarrow 2Li_2O(s) + 4NO_2(g) + O_2(g)$
$NaNO_3$, KNO_3	→ **nitrite** (nitrate(III)) + O_2 only	$2NaNO_3(s) \rightarrow 2NaNO_2(s) + O_2(g)$

- **Thermal stability increases down both groups**: $BaCO_3$ needs a much higher temperature than $MgCO_3$.
- **Group 1 compounds are more stable** than the Group 2 compounds in the same period.
- Nitrogen dioxide is a **brown, toxic, acidic** gas: decompose nitrates in a **fume cupboard**.

Why: polarisation of the anion

1. The **cation** attracts the electron cloud of the large **carbonate (or nitrate) anion** towards itself. This **polarises** (distorts) the anion.
2. The distortion **weakens the covalent C–O (or N–O) bonds inside the anion**, making it easier to break off CO_2 (or NO_2 and O_2).
3. A cation is more polarising when it has a **high charge density**: a **small ionic radius** and/or a **high charge**.
4. **Down a group**, the cation charge stays the same but the **ionic radius increases**, so charge density falls. The anion is **less polarised**, its bonds are **weakened less**, so **more energy (a higher temperature)** is needed to decompose it.
5. **Group 2 v Group 1**: M^{2+} ions have **twice the charge** and are smaller than M^+ ions in the same period, so they polarise more: Group 2 compounds decompose more easily.
6. **Lithium** is the exception in Group 1: Li^+ is very **small**, so its charge density is high enough (similar to Mg^{2+}) to polarise carbonate and nitrate ions as Group 2 ions do.

MARK-SCHEME WORDING

Standard three-mark answer (WCH12 Jun 2019, Jun 2022, Jan 2026):

- "Group 2 carbonates **increase in (thermal) stability down the group**"
- "**size of the (metal) ion increases** / charge density decreases" (charge is the same)
- "so the metal ion is **less polarising** / the (electron cloud of the) **carbonate ion is less distorted**"
- "so the **(covalent) bonds in the carbonate ion are weakened less** / more energy is needed to break them"

Must refer to **ions** (DO NOT AWARD "magnesium carbonate is less polarising"). DO NOT AWARD "weaker ionic bond". IGNORE atomic radius, effective nuclear charge. Penalise missing "ion" once only.

Nitrates: "Li and Group 2 give oxide + NO_2 + O_2; Na and K give nitrite + O_2" (WCH12 Oct 2024).

Often asked as: explain the carbonate trend; compare Li, Na and Mg nitrates (6-mark); correct the errors in a given explanation.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Lithium and magnesium behave alike (Li^+ and Mg^{2+} have similar charge densities): both nitrates give NO_2, while sodium nitrate gives nitrite and O_2 (WCH12 Oct 2024, 6 marks).

Mass-loss experiments: heat a carbonate in a weighed tube to **constant mass**, find moles of CO_2 lost, then the M_r of MCO_3 and so A_r of M (WCH12 Jan 2026). The same method finds x in a hydrated salt (WCH12 Jun 2019).

Correcting errors: "the **nitrate ion is polarised by the calcium ion**" (not the other way round) and "it weakens the covalent bonds **within** the nitrate ion" (not the bond between the ions) (WCH12 Oct 2025).

COMMON MISTAKE

Thinking $MgCO_3$ is more stable than $CaCO_3$, or that Mg^{2+} and Ca^{2+} have different charges (WCH12 Jun 2022 examiner report).

Writing "atomic radius" instead of **ionic radius**; saying "the anion is polarised" without saying by what; saying the **ionic bond** is weakened.

Talking about the stability of the **element** instead of the compound.

SPEC 8.17 Flame colours

WHAT THIS SPEC POINT ASKS

Understand how Group 1 and 2 compounds produce characteristic flame colours through electron transitions, and know the colours.

Why there is a colour

1. Heat from the flame gives energy to electrons in the metal ions. Electrons are **excited (promoted) to higher energy levels**.
2. The excited electrons are unstable and **fall back to lower energy levels** (towards the ground state).

3. As they fall, they **release the energy as light**. The energy gap is fixed for each element, so the light has **specific frequencies (wavelengths)**.
4. If those frequencies are in the **visible** region, we see a colour. Each element has its own set of energy gaps, so its own colour.

Magnesium gives **no flame colour**: its electron transitions release energy outside the visible range (in the ultraviolet), so all you see is the colour of the flame itself.

Ion	Flame colour
Li^+	**red** (crimson)
Na^+	**yellow-orange** (intense, long-lasting)
K^+	**lilac**
Rb^+	red-violet
Cs^+	blue-violet
Mg^{2+}	**none**
Ca^{2+}	**brick-red** (orange-red)
Sr^{2+}	**red** (crimson)
Ba^{2+}	**pale (apple) green**

MARK-SCHEME WORDING

- Explanation, three marks (WCH12 Oct 2025): "electrons are **excited / promoted** (by heat) to **higher energy levels**", "electrons **fall back** (to lower levels / the ground state)", "releasing (light) energy of a **specific wavelength / frequency**". IGNORE "emits a colour"; DO NOT AWARD "reflects a colour".
- Calcium: "**(brick) red / orange-red / yellow-red**"; DO NOT AWARD orange or yellow alone, scarlet or crimson (WCH12 Oct 2025).
- Potassium: "**lilac**" (ALLOW purple) (WCH12 Oct 2021).

Often asked as: the colour for a named ion; explain how a flame colour arises.

COMMON MISTAKE

Describing the atom being excited without mentioning **electrons**.

Saying "the electrons emit a colour" with no mention of energy or wavelength.

Calling calcium "orange" alone, or "crimson" (that's lithium and strontium).

SPEC 8.18 Practical procedures: thermal decomposition and flame tests

WHAT THIS SPEC POINT ASKS

Know experimental procedures to show (i) patterns in the thermal decomposition of Group 1 and 2 nitrates and carbonates (with tests for CO_2 and O_2, and recognising NO_2 by colour and acidity) and (ii) flame colours.

(i) Comparing thermal stability

CORE PRACTICAL

Suggested practical: thermal decomposition of carbonates and nitrates

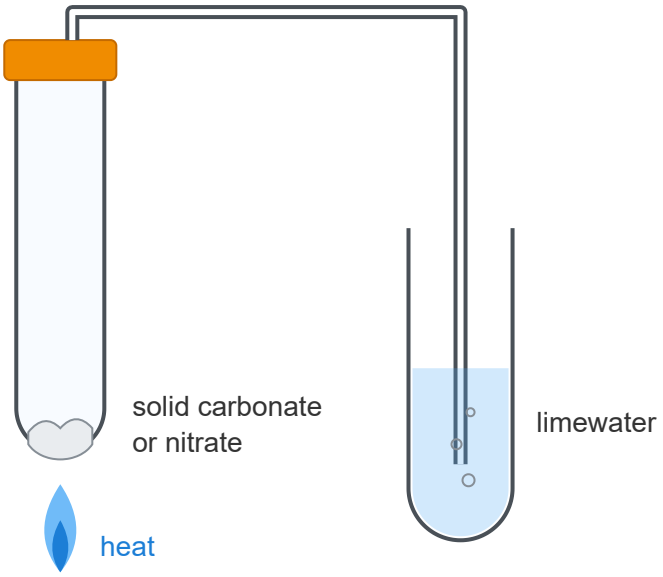
Carbonates

1. Put the **same number of moles** (e.g. 0.010 mol) of each carbonate into a boiling tube fitted with a bung and delivery tube.
2. Dip the end of the delivery tube into a fixed volume of **limewater**.
3. Heat with the **same Bunsen flame** (same height, same part of the flame).
4. Time how long it takes for the limewater to turn **milky (cloudy)**. Longer time → more stable carbonate.
5. Lift the delivery tube **out of the limewater before you stop heating**, or limewater is sucked back into the hot tube and cracks it.

Nitrates (in a fume cupboard)

1. Heat equal moles of each nitrate in a boiling tube with the same flame.
2. Time until **brown gas (NO_2)** appears (look against a white background), or test for O_2 with a **glowing splint**, which **relights**.

Safety: eye protection; NO_2 is toxic; don't heat a sealed tube.



Comparing thermal stability. Heat the same number of moles of each carbonate in the same way and time how long the limewater takes to turn cloudy. For nitrates, time how long it takes for brown NO_2 to appear, or for the gas to relight a glowing splint.

Gas	Test	Result
CO_2	bubble through **limewater** (calcium hydroxide solution)	turns **milky / cloudy** (white precipitate of $CaCO_3$)
O_2	**glowing splint**	**relights**
NO_2	look at it; damp **blue litmus**	**brown** gas; turns blue litmus **red** (acidic)

Limewater equation: **$Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$**

(ii) Flame tests

CORE PRACTICAL

Suggested practical: flame tests

1. Clean a **nichrome (or platinum) wire** loop by dipping it in **concentrated hydrochloric acid** and holding it in a **hot (blue) Bunsen flame** until it gives no colour.
2. Dip the clean wire in the acid again, then into the solid sample.
3. Hold it at the **edge of the hottest (blue) part** of the flame and note the colour.
4. Clean the wire again between samples. Sodium contamination is the usual problem: its strong yellow masks other colours.

MARK-SCHEME WORDING

- Limewater: "$CO_2(g) + Ca(OH)_2(aq) \rightarrow CaCO_3(s) + H_2O(l)$" (WCH12 Oct 2019).
- Safety: "safety glasses / goggles"; ALLOW gloves; IGNORE lab coat (WCH12 Jun 2023).
- Heating a hydrated nitrate: "magnesium nitrate **decomposes** when heated"; DO NOT AWARD "reacts with oxygen" (WCH12 Jun 2019).
- Heat a hydrated salt "**until no change in mass**" (to constant mass) (WCH12 Oct 2024).

Often asked as: tests for the gases from a decomposition; a plan for comparing stabilities or for a mass-loss experiment.

COMMON MISTAKE

Not recognising NO_2 as **acidic** (WCH12 Jan 2024 examiner report).

"Spitting" or "burning" as the explanation for mass loss from a heated nitrate (WCH12 Jun 2019 examiner report).

Using equal **masses** when the question wants a fair comparison per mole.

SPEC 8.19 Tests for carbonate, hydrogencarbonate, sulfate and ammonium ions

WHAT THIS SPEC POINT ASKS

Know the reactions and ionic equations for identifying (i) CO_3^{2-} and HCO_3^- with acid, (ii) SO_4^{2-} with acidified $BaCl_2$ and (iii) NH_4^+ with NaOH and warming.

Ion	Test	Positive result	Ionic equation
Carbonate, CO_3^{2-}	add **dilute acid** (e.g. HCl); bubble the gas through limewater	**effervescence**; limewater turns **milky**	$CO_3^{2-} + 2H^+ \rightarrow CO_2 + H_2O$
Hydrogencarbonate, HCO_3^-	add **dilute acid**; gas through limewater	**effervescence**; limewater turns **milky**	$HCO_3^- + H^+ \rightarrow CO_2 + H_2O$
Sulfate, SO_4^{2-}	add **dilute HCl** (or HNO_3), then **barium**	**white precipitate**	$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$

Ion	Test	Positive result	Ionic equation
	chloride solution		
Ammonium, NH_4^+	add **NaOH(aq)** and **warm**	gas turns damp **red litmus blue**; gives **white smoke** with HCl fumes (from a glass rod dipped in conc. HCl)	$NH_4^+ + OH^- \rightarrow NH_3 + H_2O$; then $NH_3 + HCl \rightarrow NH_4Cl$

Why the details matter

- The sulfate test is **acidified** first so that **carbonate** ions are removed (as CO_2). Otherwise $BaCO_3$, also a white precipitate, would give a false positive. Don't acidify with **sulfuric** acid: it adds sulfate ions.
- Litmus must be **damp**: ammonia acts as an alkali only when dissolved in water.
- Warming drives the ammonia out of solution.

EXAM TIP

Telling a carbonate from a hydrogencarbonate (both fizz with acid): add **magnesium sulfate** solution. A carbonate gives a white precipitate ($MgCO_3$) straight away; a hydrogencarbonate doesn't. Or heat the solid: a Group 1 hydrogencarbonate gives off CO_2 and water, but the Group 1 carbonate doesn't decompose.

MARK-SCHEME WORDING

- Carbonate + acid: "$CO_3^{2-} + 2H^+ \rightarrow CO_2 + H_2O$"; DO NOT AWARD H_2CO_3 as a product (WCH12 Oct 2019).
- $BaSO_4$: "$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$", with state symbols (WCH12 Oct 2024).
- Testing for an acid (–COOH) with a carbonate: "effervescence (and limewater turns cloudy)" (WCH12 Oct 2020).

Often asked as: name the reagent and the observation; write the ionic equation; part of an identification (Core Practical 8 style) question.

COMMON MISTAKE

Giving the reagent but not the **observation**, or the observation but not the reagent. You need both.

"Gas turns limewater cloudy" without saying a gas was **seen** (effervescence / fizzing).

SPEC 8.20 Solution concentrations and acid–base titrations

WHAT THIS SPEC POINT ASKS

Calculate concentrations in mol dm^{-3} and g dm^{-3}, including from simple acid–base titrations using methyl orange and phenolphthalein.

The equations

- **$n = c \times V$**, with V in **dm^3** ($cm^3 \div 1000$).
- **$c = n \div V$** (mol dm^{-3}).
- **Mass concentration (g dm^{-3}) = c (mol dm^{-3}) × M (g mol^{-1})**.

Indicators

Indicator	In acid	In alkali	End point	Use for
Methyl orange	**red**	**yellow**	**orange**	strong acid with strong or weak base (e.g. HCl with Na_2CO_3 or NH_3)
Phenolphthalein	**colourless**	**pink**	pale pink / just colourless	strong or weak acid with strong base (e.g. ethanoic acid with NaOH)

- For a strong acid + strong alkali, either works.
- Don't use **universal indicator**: it changes colour gradually, so there's no sharp end point.
- Describe the colour change **in the direction it happens**: alkali in the flask, acid added from the burette, phenolphthalein goes **pink** → **colourless**.

Titration calculation: the steps

1. Mean titre from the **concordant** results (within 0.10 cm^3 of each other).
2. Moles of the substance of known concentration: $n = c \times V$.
3. Use the **mole ratio** from the balanced equation.
4. Concentration of the unknown: $c = n \div V$. Scale up if a sample was diluted.

WORKED EXAMPLE

Worked example: concentration of NaOH, in mol dm^{-3} and g dm^{-3}

25.0 cm^3 of NaOH(aq) needs a mean titre of 22.40 cm^3 of 0.100 mol dm^{-3} HCl.

n(HCl) = 0.100 × 22.40 ÷ 1000 = 2.240 × 10^{-3} mol

NaOH + HCl → NaCl + H_2O (1 : 1), so n(NaOH) = 2.240 × 10^{-3} mol

c(NaOH) = 2.240 × 10^{-3} ÷ 0.0250 = **0.0896 mol dm^{-3}**

Mass concentration = 0.0896 × 40.0 = **3.58 g dm^{-3}**

WORKED EXAMPLE

Worked example: a 1 : 2 ratio

25.0 cm^3 of Na_2CO_3(aq) needs 24.00 cm^3 of 0.150 mol dm^{-3} HCl (methyl orange).

Na_2CO_3 + 2HCl → 2NaCl + CO_2 + H_2O

n(HCl) = 0.150 × 0.02400 = 3.60 × 10^{-3} mol → n(Na_2CO_3) = 1.80 × 10^{-3} mol

c(Na_2CO_3) = 1.80 × 10^{-3} ÷ 0.0250 = **0.0720 mol dm^{-3}**

WORKED EXAMPLE

Worked example: water of crystallisation

3.58 g of $Na_2CO_3 \cdot xH_2O$ is made up to 250 cm^3. 25.0 cm^3 portions need 25.00 cm^3 of 0.100 mol dm^{-3} HCl.

n(HCl) = 2.50 × 10^{-3} mol → n(Na_2CO_3) in 25.0 cm^3 = 1.25 × 10^{-3} mol

In 250 cm^3: × 10 = 1.25 × 10^{-2} mol

M of the hydrated salt = 3.58 ÷ 0.0125 = 286.4 g mol^{-1}

Mass of water per mole = 286.4 − 106.0 = 180.4 → x = 180.4 ÷ 18.0 = **10** ($Na_2CO_3 \cdot 10H_2O$)

Worked example: ethanoic acid in vinegar

1. 25.0 cm^3 of vinegar is diluted to 250 cm^3 in a volumetric flask. 25.0 cm^3 of 0.100 mol dm^{-3} NaOH needs 22.40 cm^3 of the diluted vinegar for neutralisation. Find the concentration of ethanoic acid in the original vinegar in g dm^{-3}. CH_3COOH + NaOH → CH_3COONa + H_2O.

2. Moles of sodium hydroxide:

$$n(\text{NaOH}) = 0.100 \times \frac{25.0}{1000} = 2.50 \times 10^{-3}\ \text{mol}$$

3. The reacting ratio is 1 : 1, so 2.50 × 10^{-3} mol of ethanoic acid was in 22.40 cm^3 of diluted vinegar:

$$c = \frac{2.50 \times 10^{-3}}{0.02240} = 0.1116 \text{ mol dm}^{-3}$$

4. The vinegar was diluted ten times (25.0 cm^3 made up to 250 cm^3), so the original concentration is 1.116 mol dm^{-3}.

5. Convert to g dm^{-3} using $M_r(CH_3COOH)$ = 60.0:

$$1.116 \times 60.0 = 67.0 \text{ g dm}^{-3}$$

Worked example: identifying a group 1 carbonate

1. 2.07 g of a group 1 carbonate, M_2CO_3, is dissolved and made up to 250 cm^3. 25.0 cm^3 of this solution reacts with 30.00 cm^3 of 0.100 mol dm^{-3} HCl. $M_2CO_3 + 2HCl \rightarrow 2MCl + H_2O + CO_2$. Identify M.

2. $n(\text{HCl}) = 0.100 \times 0.03000 = 3.00 \times 10^{-3}$ mol, so $n(\text{M}_2\text{CO}_3) = 1.50 \times 10^{-3}$ mol in 25.0 cm^3, and 1.50 × 10^{-2} mol in the full 250 cm^3.

3. $$M_r = \frac{2.07}{1.50 \times 10^{-2}} = 138$$

4. Subtract the carbonate (12.0 + 3 × 16.0 = 60.0) and halve: $A_r(M)$ = (138 − 60) ÷ 2 = 39, so M is **potassium** and the compound is K_2CO_3.

MARK-SCHEME WORDING

- Colour changes: "colourless to (pale) pink" (alkali added to acid) or "(pale) pink to colourless" (acid added to alkali). DO NOT AWARD red or purple; the reverse order scores only one (WCH12 Jun 2019, Oct 2021).
- Working: moles in titre → ratio → scale up (× 10 for a 250 cm^3 flask sampled with a 25.0 cm^3 pipette) → mass or M_r.
- x in a hydrated salt must be a **whole number** (WCH12 Jan 2020).
- Final answers to 2 or 3 s.f. with units.

Often asked as: concentration in both units; M_r or formula of a hydrated salt; identify a metal hydroxide; back titration.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Parts per million: e.g. mass of SO_2 in a volume of air using 24 dm³ mol⁻¹ (WCH12 Oct 2019).

Back titration to find the purity of a solid (WCH12 Oct 2025, Jun 2025).

Ideal gas equation, pV = nRT, with p in Pa and V in m³, to find an M_r (WCH12 Oct 2024).

Worked example: a back titration on an antacid tablet

1. A 0.500 g antacid tablet containing calcium carbonate is added to 50.0 cm^3 of 0.500 mol dm^{-3} hydrochloric acid (an excess). When it has reacted, the mixture is made up to 250 cm^3. 25.0 cm^3 portions need 15.20 cm^3 of 0.100 mol dm^{-3} NaOH. Find the percentage by mass of $CaCO_3$ in the tablet.

   Calcium carbonate is insoluble, so it can't be titrated directly. Instead you react it with a known excess of acid and titrate the acid that is left over.

2. Acid left in each 25.0 cm^3 portion (HCl : NaOH is 1 : 1):

$$n(\text{HCl}) = 0.100 \times \frac{15.20}{1000} = 1.52 \times 10^{-3}\ \text{mol}$$

   So the whole 250 cm^3 contained 1.52 × 10^{-2} mol of unreacted HCl.

3. Acid added at the start: $0.500 \times 0.0500 = 2.50 \times 10^{-2}$ mol. Acid used up by the tablet:

$$2.50 \times 10^{-2} - 1.52 \times 10^{-2} = 9.8 \times 10^{-3}\ \text{mol}$$

4. $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$, so the moles of $CaCO_3$ are half of this, 4.9 × 10^{-3} mol. Using $M_r(CaCO_3)$ = 100.1:

$$m = 4.9 \times 10^{-3} \times 100.1 = 0.490\ \text{g} \qquad \frac{0.490}{0.500} \times 100 = 98\%$$

COMMON MISTAKE

Forgetting the mole ratio, or using the sample volume where the titre should go (WCH12 Oct 2020 examiner report).

Not scaling up by 10 when a 25.0 cm³ portion was taken from 250 cm³.

Rounding a calculated M_r too early, so the formula comes out wrong (WCH12 Oct 2024 examiner report).

SPEC 8.21 Core Practical 3: finding the concentration of hydrochloric acid

WHAT THIS SPEC POINT ASKS

Find the concentration of a hydrochloric acid solution by titration against a standard alkali.

CORE PRACTICAL

Core practical 3: method in brief

1. If the acid is too concentrated (around 1 mol dm^{-3}), dilute it first: **pipette 25.0 cm^3** into a **250 cm^3 volumetric flask**, make up to the mark with distilled water and invert to mix. This is a tenfold dilution.
2. **Rinse** the burette with a little of the (diluted) acid, then fill it. Make sure the **jet is full** (no air bubble) and remove the funnel. Read the bottom of the meniscus at eye level, to **0.05 cm^3**.
3. **Rinse** a pipette with the standard NaOH solution (e.g. 0.100 mol dm^{-3}), then use a **pipette filler** to transfer **25.0 cm^3** into a conical flask.
4. Add 2–3 drops of indicator (methyl orange or phenolphthalein). Stand the flask on a **white tile**.
5. Run in acid while **swirling**. Do a quick **rough (rangefinder)** titration first.
6. Repeat accurately: near the end point add acid **dropwise**, rinsing the flask walls with distilled water, until **one drop** gives a permanent colour change.
7. Repeat until you have **concordant** titres (within **0.10 cm^3**). Average only the concordant ones.

WORKED EXAMPLE

Worked example: processing CP3 results

25.0 cm^3 of 0.100 mol dm^{-3} NaOH; titres of diluted HCl: rough 22.90, then 22.30, 22.50, 22.30 cm^3.

Concordant: 22.30 and 22.30 → mean **22.30 cm^3** (22.50 is not within 0.10 cm^3, so it's left out)

$n(NaOH) = 0.100 \times 0.0250 = 2.50 \times 10^{-3}$ mol = n(HCl)

$c(\text{diluted HCl}) = 2.50 \times 10^{-3} \div 0.02230 = 0.112$ mol dm^{-3}

c(original HCl) = 0.112 × 10 = **1.12 mol dm^{-3}**

Errors and why each step is there

Step	If you don't…
rinse the burette with the acid	water left inside dilutes the acid → titre too **large**
fill the jet (no air bubble)	the bubble escapes during the titration → reading too **large**
rinse the pipette with the alkali	the alkali is diluted → fewer moles → titre too **small**
add dropwise near the end point	you overshoot → titre too **large**
(water in the conical flask)	**no effect**: it doesn't change the moles of alkali

MARK-SCHEME WORDING

- Odd titres: "first titre likely to be a **rangefinder**", "**air bubble** in the burette jet", "burette **rinsed with water** (rather than the solution)"; ALLOW "overshot / not dropwise near the end point". IGNORE parallax, water in the conical flask (WCH12 Jun 2019).
- Why no third titre: "**two concordant titres** had already been obtained" (or "only 10 cm^3 of solution was left"). DO NOT AWARD "titres within ±0.1 cm^3" alone (WCH12 Oct 2019).
- Describe a titration (two marks): "known volume of one solution", "known concentration of the other", "titrate until a colour change" (WCH12 Jan 2025).

Often asked as: process titration results; explain anomalous titres; describe the method; why only two titres were needed.

COMMON MISTAKE

Describing a **rates** experiment when asked how to do a titration (WCH12 Jan 2025 examiner report: over half scored zero).

Including the rough titre in the mean.

Recording burette readings to 0.1 cm^3: always 2 decimal places, ending in 0 or 5.

SPEC 8.22 Minimising uncertainty in volumetric analysis

WHAT THIS SPEC POINT ASKS

Understand how to minimise measurement uncertainty in volumetric analysis, and estimate the overall uncertainty in a calculated result.

Uncertainty of each piece of apparatus

Percentage uncertainty = (uncertainty ÷ quantity measured) × 100

Apparatus	Typical uncertainty	Notes
2 d.p. balance	±0.005 g per reading	weighing by difference = **two** readings → ±0.01 g
25.0 cm³ pipette	±0.06 cm³	one reading
250 cm³ volumetric flask	±0.3 cm³ (class B) or less	one reading
burette	±0.05 cm³ per reading	a titre = **two** readings → ±0.10 cm³

Use the values a question gives you; these are typical.

Overall uncertainty: add the **percentage** uncertainties of every measurement used in the calculation. Then apply that percentage to the answer.

How to reduce uncertainty

- Use a **larger mass** of solid, or a **larger titre** (e.g. by changing the concentrations), so the fixed uncertainty is a smaller percentage.
- Use more precise apparatus: a 3 d.p. balance; a pipette and volumetric flask rather than measuring cylinders.
- Repeat to get **concordant** titres and take a mean (this reduces random error, not the apparatus uncertainty).
- Avoid systematic errors: rinse apparatus with the solution it will hold, read the **bottom of the meniscus at eye level**, remove the funnel from the burette, fill the jet.

WORKED EXAMPLE

Worked example: overall uncertainty (CP4 data)

Mass by difference 2.45 g (±0.01 g): 0.01 ÷ 2.45 × 100 = 0.41%

250 cm³ flask (±0.3 cm³): 0.3 ÷ 250 × 100 = 0.12%

25.0 cm³ pipette (±0.06 cm³): 0.06 ÷ 25.0 × 100 = 0.24%

Titre 24.80 cm³ (±0.10 cm³): 0.10 ÷ 24.80 × 100 = 0.40%

Total ≈ **1.2%**. For c(NaOH) = 0.102 mol dm^{-3}: 1.2% of 0.102 = ±0.001, so 0.102 ± 0.001 mol dm^{-3}.

MARK-SCHEME WORDING

- Percentage uncertainty for a quantity from two readings: "(2 × 0.05 × 100) ÷ 13.2 = 0.76%". DO NOT AWARD 0.75 or 0.80 (WCH12 Jun 2022).

Often asked as: calculate a percentage uncertainty; suggest how to reduce it; choose suitable apparatus.

COMMON MISTAKE

Using 0.05 cm³ for a titre: a titre is the **difference** of two readings, so ±0.10 cm³.

Thinking repeating the titration reduces the uncertainty of the burette itself. It improves reliability, not the apparatus's precision.

SPEC 8.23 Core Practical 4: a standard solution from a solid acid

WHAT THIS SPEC POINT ASKS

Make a standard solution of a solid acid and use it to find the concentration of a sodium hydroxide solution.

What a standard solution is

A **standard solution** has an **accurately known concentration**. You make it by dissolving an accurately weighed mass of a pure, stable solid in water and making it up to an exact volume. A suitable solid acid is **sulfamic acid**, H_2NSO_3H (M_r 97.1), which is monoprotic: $H_2NSO_3H + NaOH \rightarrow H_2NSO_3Na + H_2O$ (1 : 1).

CORE PRACTICAL

Core practical 4: method in brief

Making the standard solution

1. Weigh a weighing bottle containing about 2.5 g of the solid acid on a 2 d.p. (or 3 d.p.) balance.
2. Tip the solid into a **beaker**, then **reweigh** the bottle: mass used = difference (**weighing by difference**, so solid left in the bottle doesn't matter).
3. Dissolve the solid in about 100 cm³ of distilled water, stirring with a glass rod.
4. Pour the solution through a funnel into a **250 cm³ volumetric flask**. **Rinse** the beaker, rod and funnel with distilled water and add all the **rinsings** to the flask.
5. Add distilled water until the **bottom of the meniscus** is on the line (use a dropping pipette for the last few drops, read at eye level).
6. Stopper and **invert several times** to mix.

Titration

7. Pipette 25.0 cm^3 of the standard acid into a conical flask; add phenolphthalein.
8. Titrate with the NaOH from the burette until the solution turns **pale pink** (colourless → pink). Repeat to concordant titres.

WORKED EXAMPLE

Worked example: processing CP4 results

Mass of sulfamic acid = 2.45 g → n = 2.45 ÷ 97.1 = 0.02523 mol in 250 cm^3

c(acid) = 0.02523 ÷ 0.250 = 0.1009 mol dm^{-3}

In 25.0 cm^3: n = 0.1009 × 0.0250 = 2.523 × 10^{-3} mol = n(NaOH) (1 : 1)

Mean titre of NaOH = 24.80 cm^3

c(NaOH) = 2.523 × 10^{-3} ÷ 0.02480 = **0.102 mol dm^{-3}**

Why each step matters

- **Rinsings** go into the flask so that **all** the weighed acid ends up in the solution.
- **Inverting** makes the concentration the same throughout: otherwise each 25.0 cm^3 portion would have a different amount of acid.
- Overfilling past the line can't be undone: the solution must be remade.
- NaOH can't be used as a standard itself: the solid absorbs **water** and **CO_2** from the air, so its mass isn't a reliable measure of the moles of NaOH. That's why its concentration has to be found by titration.

MARK-SCHEME WORDING

- The calculation is what gets marked: moles of solid in the 250 cm^3 flask, ÷ 10 for each 25.0 cm^3 portion (or × 10 going the other way), then the mole ratio (WCH12 Jan 2020, a 250 cm^3 standard solution of a carbonate).
- No paper from 2019 to 2026 asked for the full preparation method, so learn it from the practical itself: weigh by difference, dissolve, transfer **with rinsings**, make up to the mark, invert.

Often asked as: calculate the concentration of the standard solution and then of the NaOH; explain a step in making the solution.

COMMON MISTAKE

Forgetting to add the rinsings, or making the solution up to the mark **before** the solid has fully dissolved.

Using the mass in the whole 250 cm^3 for the moles in one 25.0 cm^3 portion.

SPEC 8.24 Group 7 trends: physical properties, electronegativity and reactivity

WHAT THIS SPEC POINT ASKS

Understand the reasons for the trends in Group 7 (chlorine, bromine, iodine) in (i) melting and boiling temperatures and state, (ii) electronegativity and (iii) reactivity.

(i) Melting and boiling temperatures, and state

	F_2	Cl_2	Br_2	I_2
State and colour at room temperature	pale yellow gas	**pale green** gas	**red-brown liquid** (gives orange-brown vapour)	**grey-black (shiny) solid**; gives a **purple vapour** when warmed
Boiling temperature / °C	−188	−34	59	184
Electrons per molecule	18	34	70	106

- Melting and boiling temperatures **increase down the group**.
- The halogens are simple molecules, X_2, held together by **London forces only**.
- Down the group there are **more electrons per molecule**, so the **London forces get stronger**, and **more energy is needed to separate the molecules**. So the state goes gas → liquid → solid.

(ii) Electronegativity

Electronegativity is the **ability of an atom to attract the bonding pair of electrons in a covalent bond**.

	F	Cl	Br	I
Electronegativity (Pauling)	4.0	3.0	2.8	2.5

Electronegativity **decreases down the group**: the atoms have **more shells**, so the bonding pair is **further from the nucleus** and **more shielded** by inner electrons. It is less strongly attracted, even though the nuclear charge is higher.

(iii) Reactivity

- Halogens react by **gaining an electron** to form X^- ions (or by attracting a shared pair). They are **oxidising agents**.

- Down the group, the incoming electron goes into a shell **further from the nucleus** with **more shielding**, so it is **attracted less strongly**.
- So **reactivity (oxidising power) decreases down the group**: $F_2 > Cl_2 > Br_2 > I_2$.

MARK-SCHEME WORDING

- Iodine: "**(dark) grey / black** and **solid**". Both state and colour are needed; IGNORE shiny (WCH12 Jan 2022).
- Electronegativity definition: "the **ability of an atom to attract a bonding / shared pair of electrons** (in a covalent bond)" (WCH12 Jan 2022).
- Trend (two marks): "elements become **less electronegative** down the group"; "the outer electrons are **further from the nucleus / more shielding**, so the bonding pair is **less attracted**". DO NOT AWARD "nuclear charge decreases". IGNORE ionic radius. One paper ignored "atomic radius increases" on its own, another accepted it, so give the full chain (WCH12 Jan 2022, Jun 2022).

Often asked as: state and colour of a halogen; define electronegativity; explain the electronegativity or boiling-temperature trend.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Electronegativity across a period: bromine has one more proton than selenium in the same shell, so it attracts a bonding pair more strongly; compared with chlorine, bromine has an extra shell, so it attracts less strongly (WCH12 Jan 2026).

COMMON MISTAKE

Confusing electronegativity with electron affinity or ionisation energy (WCH12 Jun 2022 examiner report).

Writing about the size of the **ion** in an electronegativity answer.

Missing "pair of electrons" or "covalent bond" from the definition.

Saying the covalent X–X bonds break when a halogen boils.

SPEC 8.25 Halogen displacement reactions and colours

WHAT THIS SPEC POINT ASKS

Understand the trend in reactivity of Cl_2, Br_2 and I_2 through their redox reactions with halide ions in solution, and know the colours of the halogens as elements, in water and in a non-polar solvent.

Displacement

A **more reactive halogen displaces** (oxidises) the ion of a **less reactive** halogen from solution.

Added halogen	$Cl^-(aq)$	$Br^-(aq)$	$I^-(aq)$
$Cl_2(aq)$	—	**yes**: yellow/orange	**yes**: brown
$Br_2(aq)$	no reaction	—	**yes**: brown
$I_2(aq)$	no reaction	no reaction	—

- $Cl_2(aq) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$
- $Cl_2(aq) + 2I^-(aq) \rightarrow 2Cl^-(aq) + I_2(aq)$
- $Br_2(aq) + 2I^-(aq) \rightarrow 2Br^-(aq) + I_2(aq)$

In each: the halogen is **reduced** (0 → −1) and acts as the oxidising agent; the halide ion is **oxidised** (−1 → 0). The metal ion (K^+, Na^+) is a spectator.

This shows **oxidising power decreases down the group** ($Cl_2 > Br_2 > I_2$), and so the **reducing power of the halide ions increases** ($I^- > Br^- > Cl^-$).

Colours you must know

Halogen	As the element	In water	In a non-polar solvent (cyclohexane, hexane)
Cl_2	pale green gas	very pale green (almost colourless)	very pale green (almost colourless)
Br_2	red-brown liquid	**yellow / orange**	**orange (red-orange)**
I_2	grey-black solid	**brown** (pale yellow when very dilute)	**purple / violet / pink**

Brown in water could be bromine or iodine. To tell them apart, **add cyclohexane and shake**: the halogen moves into the upper organic layer, which is **orange** for bromine and **purple** for iodine.

MARK-SCHEME WORDING

- "$Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$" (ALLOW halves); DO NOT AWARD uncancelled K^+ (WCH12 Jun 2025).
- Chlorine added to bromide: the solution turns "**orange / yellow**"; DO NOT AWARD brown or green (WCH12 Jan 2026).
- Iodine shaken with water and cyclohexane: "(pale) yellow aqueous layer", "pink / purple / violet cyclohexane layer". DO NOT AWARD brown alone, colourless, red or grey (WCH12 Jun 2019).

Often asked as: equation and colour change for a displacement; colours of the layers; which halide is oxidised first.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Competitive displacement: when chlorine is added to a mixture of iodide and bromide, **iodide is oxidised first** (it is the stronger reducing agent). E.g. 0.200 mol Cl_2 with 0.300 mol I^- and 0.300 mol Br^-: 0.150 mol Cl_2 makes 0.150 mol I_2, and the remaining 0.050 mol Cl_2 makes 0.050 mol Br_2 (WCH12 Jan 2026 style).

COMMON MISTAKE

Saying "chlorine displaces bromine" when the reactant is bromide: chlorine displaces bromine **from bromide ions**.

Calling the aqueous layer colourless when only "most" of the iodine moved to the organic layer (WCH12 Jun 2019 examiner report).

Writing a reaction for $Br_2 + Cl^-$ or $I_2 + Br^-$: there isn't one.

SPEC 8.26 Redox reactions of the halogens

WHAT THIS SPEC POINT ASKS

Understand, in terms of oxidation number changes: (i) oxidation of Group 1 and 2 metals; (ii) disproportionation of chlorine with water and its use in water treatment; (iii) with cold, dilute NaOH to make bleach; (iv) with hot alkali; (v) analogous reactions.

(i) With Group 1 and 2 metals

The halogen is reduced (0 → −1); the metal is oxidised (0 → +1 or +2):

- $2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$
- $Ca(s) + Cl_2(g) \rightarrow CaCl_2(s)$
- $Mg(s) + Br_2(l) \rightarrow MgBr_2(s)$
- $2K(s) + I_2(s) \rightarrow 2KI(s)$

Reactions get less vigorous with the halogen going down Group 7, and more vigorous with the metal going down Groups 1 and 2.

(ii) Chlorine with water; water treatment

$Cl_2(g) + H_2O(l) \rightleftharpoons HCl(aq) + HClO(aq)$

- Chlorine: 0 → **−1** in HCl (reduced) and 0 → **+1** in HClO, chloric(I) acid (oxidised). A **disproportionation**.
- **HClO** (and the ClO^- ion it forms) **kills bacteria** and other microorganisms, which is why chlorine is added to drinking water and swimming pools.
- **Benefits**: prevents waterborne diseases such as cholera and typhoid, and the chlorine stays in the water to protect it in the pipes.
- **Risks**: chlorine is **toxic**; it can react with organic matter in water to form **chlorinated organic compounds** (e.g. chloroalkanes) that may cause cancer. The benefit to public health is judged to far outweigh the risk, at low concentrations.
- Adding **alkali** removes H^+ (from HCl and HClO), so the equilibrium shifts **to the right**.

(iii) Chlorine with cold, dilute sodium hydroxide: bleach

$Cl_2(g) + 2NaOH(aq) \rightarrow NaCl(aq) + NaClO(aq) + H_2O(l)$ (about 15 °C)
Ionic: $Cl_2 + 2OH^- \rightarrow Cl^- + ClO^- + H_2O$

- Chlorine: 0 → **−1** (Cl^-, reduced) and 0 → **+1** (ClO^-, chlorate(I), oxidised).
- The solution of **sodium chlorate(I)** (sodium hypochlorite) is household **bleach**; ClO^- is the active ingredient.

(iv) Chlorine with hot alkali

$3Cl_2(g) + 6NaOH(aq) \rightarrow 5NaCl(aq) + NaClO_3(aq) + 3H_2O(l)$ (about 70 °C)
Ionic: $3Cl_2 + 6OH^- \rightarrow 5Cl^- + ClO_3^- + 3H_2O$

- Chlorine: 0 → **−1** (five Cl atoms, reduced) and 0 → **+5** (one Cl atom in ClO_3^-, chlorate(V), oxidised). Electrons balance: 5 × 1 gained = 1 × 5 lost.
- In hot alkali, chlorate(I) ions themselves disproportionate: $3ClO^- \rightarrow 2Cl^- + ClO_3^-$.

(v) Analogous reactions

The same patterns work with other alkalis and halogens. Work out the products by analogy and check the oxidation numbers.

- Bromine, cold: $Br_2 + 2NaOH \rightarrow NaBr + NaBrO + H_2O$. Hot: $3Br_2 + 6NaOH \rightarrow 5NaBr + NaBrO_3 + 3H_2O$.
- Iodine behaves similarly; iodate(I) disproportionates very readily, so IO_3^- forms even when cold.
- Calcium hydroxide, cold: $2Cl_2 + 2Ca(OH)_2 \rightarrow CaCl_2 + Ca(ClO)_2 + 2H_2O$ (bleaching powder).
- Calcium hydroxide, hot: $6Cl_2 + 6Ca(OH)_2 \rightarrow 5CaCl_2 + Ca(ClO_3)_2 + 6H_2O$.

WORKED EXAMPLE

Worked example: an unfamiliar disproportionation

$3Ca(OH)_2 + 3Cl_2 + KCl \rightarrow KClO_3 + 3CaCl_2 + 3H_2O$ (a route to potassium chlorate(V))

Chlorine in Cl_2: 0. In $KClO_3$: K +1, O 3 × (−2) = −6, so Cl = **+5** (oxidised). In $CaCl_2$: **−1** (reduced).

Same element, one species (Cl_2), oxidised and reduced: **disproportionation**. Calcium stays +2, so don't mention it.

MARK-SCHEME WORDING

- Water: "$Cl_2 + H_2O \rightleftharpoons HCl + HOCl$" (ALLOW HClO, $H^+ + Cl^-$ + HOCl) (WCH12 Jan 2022).
- Cold: "$Cl_2 + 2NaOH \rightarrow NaClO + NaCl + H_2O$"; hot: "$3Cl_2 + 6NaOH \rightarrow NaClO_3 + 5NaCl + 3H_2O$". Ionic forms allowed. Cl oxidised to **+1, +1 and +5** respectively and reduced to **−1** in all three; name **disproportionation** (WCH12 Jan 2022, 6 marks).
- Alkali on the equilibrium: NaOH "reacts with / removes the HCl / H^+", so the position shifts "to the right" (WCH12 Oct 2024).
- Acid on bleach releases chlorine: "$NaClO + 2HCl \rightarrow NaCl + H_2O + Cl_2$"; DO NOT AWARD $NaClO + HCl \rightarrow NaOH + Cl_2$ (WCH12 Jun 2023).
- "$3Ca(OH)_2 + 3Cl_2 + KCl \rightarrow KClO_3 + 3CaCl_2 + 3H_2O$" (WCH12 Jun 2021).

Often asked as: equations and oxidation-number changes for the three chlorine reactions; benefits and risks of chlorinating water; why mixing acid with bleach is dangerous.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Atom economy of making a chlorate: e.g. about 24% for the $Ca(OH)_2$/KCl route (WCH12 Jun 2021, Oct 2025).

Chlorates(V) and (VII) and chlorine dioxide, ClO_2 (an unpaired electron), in an unfamiliar context (WCH12 Jun 2024).

COMMON MISTAKE

Showing an oxidation-number change for Ca or Na (WCH12 Jan 2023 examiner report).

Writing a sulfuric acid reaction when the question says "alkali" (WCH12 Jan 2022 examiner report).

Mixing up which conditions give ClO^- (cold, dilute) and which give ClO_3^- (hot).

SPEC 8.27 Reactions of halides and hydrogen halides

WHAT THIS SPEC POINT ASKS

Understand: (i) solid Group 1 halides with concentrated sulfuric acid, showing the trend in reducing ability of the hydrogen halides; (ii) precipitation of Cl^-, Br^- and I^- with acidified silver nitrate, and the solubility of the precipitates in ammonia; (iii) hydrogen halides with ammonia gas and with water.

(i) Solid halides with concentrated sulfuric acid

All three first make the hydrogen halide (an **acid–base** reaction, not redox):
$NaX(s) + H_2SO_4(l) \rightarrow NaHSO_4(s) + HX(g) \rightarrow$ **misty (steamy) fumes**

Then HBr and HI, which are stronger **reducing agents**, go on to **reduce the sulfuric acid**:

Halide	Further reaction	Sulfur product (oxidation number)	What you see
NaCl	none: HCl can't reduce H_2SO_4	— (stays +6)	**misty fumes** of HCl only
NaBr	$2HBr + H_2SO_4 \rightarrow Br_2 + SO_2 + 2H_2O$	**SO_2 (+4)**	misty fumes; **red-brown / orange fumes** of Br_2; colourless choking gas (SO_2)
NaI	$2HI + H_2SO_4 \rightarrow I_2 + SO_2 + 2H_2O$	SO_2 (+4)	misty fumes; **purple vapour / black solid** (I_2)
	$6HI + H_2SO_4 \rightarrow 3I_2 + S + 4H_2O$	**S (0)**	**yellow solid**
	$8HI + H_2SO_4 \rightarrow 4I_2 + H_2S + 4H_2O$	**H_2S (−2)**	**rotten-egg smell**

The trend: reducing power **increases** $Cl^- < Br^- < I^-$ (HCl < HBr < HI). The bigger the halide ion, the further its outer electrons are from the nucleus and the more shielded they are, so they are **lost more easily**. Iodide reduces sulfur all the way from +6 to −2; bromide only to +4; chloride not at all.

Iodide ionic equation (no spectators): **$8I^- + 10H^+ + SO_4^{2-} \rightarrow 4I_2 + H_2S + 4H_2O$** (or $8I^- + 8H^+ + H_2SO_4 \rightarrow 4I_2 + H_2S + 4H_2O$).

Iodide ions reducing sulfuric acid to hydrogen sulfide

1. Oxidation: $2I^- \rightarrow I_2 + 2e^-$. Reduction: $SO_4^{2-} + 10H^+ + 8e^- \rightarrow H_2S + 4H_2O$.

2 Multiply the iodide half-equation by 4 to release 8 electrons, then add:

$$8I^- + SO_4^{2-} + 10H^+ \rightarrow 4I_2 + H_2S + 4H_2O$$

Charges: left = −8 − 2 + 10 = 0, right = 0.

(ii) Silver nitrate test for halide ions

1. Add **dilute nitric acid** to the solution. This removes ions such as CO_3^{2-} and OH^- that would also give a precipitate with Ag^+. (Not HCl, which adds Cl^-; not H_2SO_4.)
2. Add **silver nitrate solution**.
3. Then test the precipitate's solubility in **ammonia solution**.

Ion	Precipitate	Ionic equation	Dilute $NH_3(aq)$	Concentrated $NH_3(aq)$
Cl^-	**white** AgCl	$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$	**dissolves**	dissolves
Br^-	**cream** AgBr	$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$	doesn't dissolve	**dissolves**
I^-	**(pale) yellow** AgI	$Ag^+(aq) + I^-(aq) \rightarrow AgI(s)$	doesn't dissolve	**doesn't dissolve**

The colours are hard to tell apart (white v cream v pale yellow), which is why the ammonia step is needed. The precipitate dissolves because ammonia forms the soluble complex ion $[Ag(NH_3)_2]^+$. Fluoride gives **no precipitate**: AgF is soluble.

(iii) Hydrogen halides with ammonia and with water

- With **ammonia gas**: a **white smoke** of the ammonium halide forms. **$HCl(g) + NH_3(g) \rightarrow NH_4Cl(s)$**. This is the test for HCl fumes (and for ammonia).
- With **water**: hydrogen halides are very soluble and react to form **acidic solutions**: **$HCl(g) + H_2O(l) \rightarrow H_3O^+(aq) + Cl^-(aq)$** (hydrochloric acid). HBr and HI give hydrobromic and hydroiodic acid. The "misty fumes" are droplets of acid formed with moisture in the air.

MARK-SCHEME WORDING

- Iodide + conc. H_2SO_4 (WCH12 Oct 2025, 6 marks): "misty fumes (white smoke with NH_3) = HI", "purple vapour = I_2", "rotten-egg smell = H_2S", "I: −1 → 0; S: +6 → −2", "chloride doesn't change the oxidation number of S, so iodide is the stronger reducing agent", "$8I^- + 8H^+ + H_2SO_4 \rightarrow 4I_2 + H_2S + 4H_2O$". DO NOT AWARD SO_2 when the stem mentions a rotten-egg smell.

- KCl v KBr (WCH12 Jan 2020): similarities "both produce hydrogen halides / misty fumes" (acid–base, protonation); differences "**only the KBr reaction is redox**", "bromide is a stronger reducing agent", "(red-)brown fumes / SO_2". DO NOT AWARD white smoke, rotten-egg smell or yellow solid for bromide.
- Lowest-oxidation-state sulfur **compound**: SO_2 from HBr, **H_2S** from HI (sulfur itself is not a compound) (WCH12 Jun 2022).
- Silver nitrate: "add (dilute) **nitric acid** and silver nitrate", "white precipitate", "dissolves in **dilute** ammonia" (WCH12 Jun 2021). Bromide: "cream / off-white precipitate", "$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$", dissolves in **concentrated** ammonia (WCH12 Jan 2026). DO NOT AWARD HCl or H_2SO_4 as the acid.

Often asked as: observations and equations for a halide with conc. H_2SO_4; the silver nitrate test; explain the trend in reducing power.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

White smoke forms only where the fumes meet ammonia: use it to identify HX fumes (WCH12 Jun 2019).

The silver nitrate test after hydrolysing halogenoalkanes: **yellow** precipitate for an iodoalkane, **white** for a chloroalkane (WCH12 Oct 2024; see 10.10).

COMMON MISTAKE

Mixing up the silver nitrate test with Tollens' reagent and adding NaOH (WCH12 Jan 2021 examiner report).

Confusing hydrogen halides (HX) with halogens (X_2), or listing sulfur as a "compound".

Saying chloride ions reduce sulfuric acid.

SPEC 8.28 Predicting the properties of fluorine and astatine

WHAT THIS SPEC POINT ASKS

Use the trends in Group 7 to predict the properties of fluorine and astatine and their compounds.

Extrapolate every trend

Property	Trend down the group	**Fluorine** (top)	**Astatine** (bottom)
State and colour	darker, gas → solid	pale yellow **gas**	**black solid**

Property	Trend down the group	**Fluorine** (top)	**Astatine** (bottom)
Boiling temperature	increases (more electrons, stronger London forces)	lowest (−188 °C)	highest
Electronegativity	decreases	**highest of all elements (4.0)**	lowest in the group
Oxidising power of X_2	decreases	**strongest** oxidising agent	weakest
Displacement	each halogen displaces those below it	displaces all other halogens from their halides	displaces none; is displaced by all the others
Reducing power of X^-	increases	F^- the **weakest** reducing agent	At^- the **strongest** reducing agent
X^- + conc. H_2SO_4	more reduction of S	HF only (misty fumes), no redox	expected to reduce S to H_2S, like or beyond iodide
Silver halide	colour darkens, solubility in NH_3 falls	**AgF is soluble**: no precipitate	AgAt: a precipitate darker than AgI, insoluble in concentrated NH_3
HX thermal stability	decreases (H–X bond weaker)	HF very stable	HAt very unstable

Two predictions worth writing out

- Fluorine is so strong an oxidising agent that it **oxidises water**: $2F_2(g) + 2H_2O(l) \rightarrow 4HF(aq) + O_2(g)$. (Chlorine only reacts reversibly with water; see 8.26.)
- Astatine is predicted to react with hydrogen only slowly and reversibly, like iodine, and to form HAt, which, by analogy with HI, would dissolve in water to give an acid: $HAt(g) + H_2O(l) \rightarrow H_3O^+(aq) + At^-(aq)$.

EXAM TIP

For any "predict" question, state the **prediction**, then the **trend** you used ("because oxidising power decreases down the group"), then the **reason** for the trend (more shells, more shielding, weaker attraction for an incoming electron).

MARK-SCHEME WORDING

- No WCH12 paper from 2019 to 2026 asked about fluorine or astatine directly. Expect them in multiple-choice questions or as a "predict" part using the trends from 8.24 to 8.27.

Often asked as: predict the state, colour, or a reaction of astatine or fluorine; predict the result of the silver nitrate test with fluoride or astatide.

COMMON MISTAKE

Predicting a precipitate with $AgNO_3$ for fluoride. AgF is soluble.

Predicting that astatine is a stronger oxidising agent than iodine: the trend runs the other way.

Quick checks

QUICK CHECK

What is the oxidation number of chlorine in $NaClO_3$?

- −1
- +3
- +5
- +7

Sodium is +1 and the three oxygens are −6 in total, so chlorine must be +5 for the compound to be neutral: (+1) + x + (−6) = 0. That is why $NaClO_3$ is called sodium chlorate(V). Chlorine is only −1 when it is not combined with oxygen or fluorine.

QUICK CHECK

In the reaction $3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$, what happens to the chlorine?

- It is both oxidised and reduced.
- It is oxidised only.
- It is reduced only.
- It is neither oxidised nor reduced.

Chlorine starts at 0. In NaCl it is −1 (reduced) and in $NaClO_3$ it is +5 (oxidised). One element in one species going both ways is disproportionation.

QUICK CHECK

Which statement explains why the first ionisation energy decreases down group 2?

- The nuclear charge decreases down the group.
- The outer electrons are in s orbitals.
- There are fewer inner electrons to shield the outer electrons.
- The outer electrons are further from the nucleus and more shielded, which outweighs the greater nuclear charge.

Nuclear charge actually increases down the group, but each element has an extra shell, so the outer electrons are further away and shielded by more inner electrons. The attraction to the nucleus is weaker overall.

QUICK CHECK

Which group 2 carbonate has the highest decomposition temperature, and why?

- $MgCO_3$, because Mg^{2+} has the highest charge density.
- $BaCO_3$, because Ba^{2+} has the lowest charge density and polarises the carbonate ion least.
- $BaCO_3$, because barium is the most reactive metal.
- $MgCO_3$, because its lattice is the strongest.

The large Ba^{2+} ion distorts the carbonate ion's electron cloud least, so the C–O bonds are weakened least and more heat is needed to break the ion up. Mg^{2+} has the highest charge density, so $MgCO_3$ decomposes most easily.

QUICK CHECK

Concentrated sulfuric acid is added to solid sodium bromide. Which products form?

- HBr only
- HBr, Br_2, SO_2 and H_2S
- HBr, Br_2 and SO_2
- Br_2 and H_2S only

Bromide is a strong enough reducing agent to reduce sulfur from +6 to +4 (SO_2), but not as far as H_2S (−2). Only iodide, the strongest reducing agent, gets that far. HBr forms first in the acid–base step.

QUICK CHECK

A solution gives a cream precipitate with acidified silver nitrate. The precipitate dissolves in concentrated ammonia but not in dilute ammonia. Which ion is present?

- Cl^-
- Br^-
- I^-
- F^-

Silver bromide is cream and dissolves only in concentrated ammonia. Silver chloride is white and dissolves in dilute ammonia; silver iodide is pale yellow and insoluble in both. Fluoride gives no precipitate.

QUICK CHECK

A burette reading has an uncertainty of ±0.05 cm^3. What is the percentage uncertainty in a titre of 12.50 cm^3?

- 0.4%
- 0.8%
- 0.05%
- 8%

A titre uses two readings, so its uncertainty is ±0.10 cm^3: 0.10 ÷ 12.50 × 100 = 0.8%. Using only one reading's uncertainty gives 0.4%.